

CBSC-ZDC: An Auditable Specification for a Four-Momentum-Conditioned Fast Monte Carlo Surrogate of Neutron Showers in a Zero Degree Calorimeter

A mathematical research plan, implementation contract, and falsifiable comparison protocol

ASIoP ZDC FastMC Project
Independent research specification

23 July 2026

Abstract

This document specifies a reproducible research experiment for learning the readout-level distribution of single-neutron Geant4 showers in a fixed Zero Degree Calorimeter (ZDC). The sole raw per-event condition is the incident four-momentum $p^\mu = (E, p_x, p_y, p_z)$; all other event features must be deterministic functions of that vector, while detector geometry is frozen metadata. The proposed model, called the Constrained Budgeted Stochastic Cascade for ZDCs (CBSC-ZDC), is deliberately assembled from established components: a response hurdle, a bounded or target-correct response distribution, a first-visible-layer survival model, an exact-zero longitudinal support model, a masked-simplex energy profile, a finite-support count model, conditional flow matching, edge-conditioned graph message passing, causal layer attention, Gumbel-Top- k support sampling, and exact budget-preserving decoding. No individual component is claimed as a new architecture.

The central correction to the motivating idea is that deposited energy must *not* be forced to decrease from layer to layer. Neutron-induced hadronic showers may begin late, grow, fluctuate, and exhibit secondary maxima. Only the remaining accounting budget is required to be non-increasing. A second central contribution is an exact anti-dust decoder: a generated layer count selects exactly that many cells, all unselected cells are exactly zero, and selected cells receive a threshold floor plus a normalized share of the remaining layer budget. This prevents a model from improving an energy loss by adding a small positive value to every nominally empty cell.

The document is an auditor-ready design specification, not a report of validated physics performance. If the stated data assumptions hold and the implementation follows the mathematical contract, the resulting experiment should be structurally valid and reproducible, with explicit mechanisms and diagnostics designed to prevent or reveal the specific support, energy-accounting, graph, and exposure-bias defects identified in earlier project models. No architecture can guarantee agreement with Geant4 before training and held-out evaluation; that question remains empirical.

Assurance statement. “Implementing this specification exactly should work” means that the experiment should be scientifically interpretable and should satisfy the proved algebraic invariants under the stated target assumptions. It does *not* mean that the trained neural distribution is guaranteed to reproduce Geant4 within any numerical tolerance. Fidelity, speed, and usefulness must be measured.

Contents

1	Scope, status, and precise claim	4
1.1	Research question	4
1.2	What is and is not claimed	4
1.3	Relationship to previous project models	4
2	Physics basis: neutrons, hadronic showers, ZDCs, and Geant4	4
2.1	Why neutrons make the longitudinal problem stochastic	4
2.2	Role of a Zero Degree Calorimeter	5
2.3	Why the model should emulate the scored target, not microscopic transport	5
3	Data contract and supported condition manifold	5
3.1	Sole raw condition	5
3.2	Static detector metadata	6
3.3	Entry position derived from four-momentum	6
3.4	Target mode must be frozen before training	6
3.5	Energy support assumption	6
4	Statistical target and factorization	7
5	Longitudinal cascade model	7
5.1	Visible-response hurdle	7
5.2	Total response	8
5.3	First-visible-layer survival model	8
5.4	Exact active-layer support	8
5.5	Masked-simplex energy allocation	8
6	Finite-support count model	9
7	Parallel conditional flow-matching spatial model	9
7.1	Why parallel rather than a 65-call rollout	9
7.2	Continuous target state	9
7.3	Flow-matching objective	10
7.4	Velocity network	10
7.5	Sampling	10
8	Exact sparsity and the anti-dust decoder	11
8.1	The failure mode	11
8.2	Primary exact-support construction	11
8.3	What exact Top- k does not guarantee	12
8.4	Training the discrete support	12
9	Training objectives	12
10	Data pipeline and provenance	13
10.1	Raw ROOT audit	13
10.2	Canonical tensors	13
10.3	Splitting	13
10.4	Training range experiment	14

11 Staged implementation and Vertex AI execution plan	14
11.1 Stage sequence	14
11.2 Vertex job layout	15
11.3 Precision and performance	15
12 Baselines and controlled comparisons	15
12.1 Direct ZDC references	15
12.2 Internal matched baseline ladder	15
13 Recommended QA and reporting	16
13.1 Software and algebraic QA	16
13.2 Training-time monitoring	17
13.3 Distributional evaluation	17
13.4 Downstream and amplification studies	17
14 Failure modes and required interpretations	18
15 Implementation contract	18
15.1 Reference tensor interface	18
15.2 Reference hyperparameters for the first pilot	19
15.3 Pseudocode	19
16 Software QA status of the supplied scaffold	20
17 Research literature audit and evidence depth	20
18 Five-minute explanation for a general science-fair judge	21
19 Five-minute explanation for a computational HEP physicist	21
20 Conclusions	22
A Claim-to-evidence map	22
B Chronological design corrections	23
C Recommended auditor checklist	23

1 Scope, status, and precise claim

1.1 Research question

Let $Y \in \mathbb{R}_{\geq 0}^{6790}$ denote the event-level readout vector for one neutron traversing the fixed detector geometry $\mathcal{G}$. The research question is

$$\text{Can } p_{\theta}(Y | p^{\mu}, \mathcal{G}) \text{ approximate } p_{\text{G4}}(Y | p^{\mu}, \mathcal{G}) \quad (1)$$

throughout the declared 50–250 GeV reporting domain, while remaining computationally useful and respecting the exact readout contract?

The available simulation support is described as approximately 0–300 GeV non-pencil-beam neutron Geant4 data. The broader range may be used for a controlled training-support comparison, but primary results must remain separated from lower- and upper-energy stress regions.

1.2 What is and is not claimed

This specification claims the following.

- C1.** The original monotone-per-layer-deposit constraint is physically inappropriate for hadronic neutron showers; a monotone remaining budget is the correct conservation structure.
- C2.** The proposed factorization uses known, published model classes and can be implemented without inventing a new primitive architecture.
- C3.** Under explicit target assumptions, the decoder guarantees nonnegative cell outputs, exact zeros outside generated support, threshold consistency, and exact accounting identities.
- C4.** The proposed training and reporting protocol separates implementation validity from empirical detector fidelity.

It does not claim that CBSC-ZDC has already been trained, that it is superior to existing ZDC methods, or that its output is experimentally calibrated detector data. Geant4 itself is a configured stochastic transport simulation whose predictions depend on the detector, materials, production cuts, physics list, and scoring definition [1, 2, 3, 4].

1.3 Relationship to previous project models

Two prior trained models provide negative evidence and reusable engineering lessons. The legacy recurrent HurdleGraph model exhibited an unbounded total-response tail, positive energy in nominally inactive layers, invalid cross-layer slot correspondence, unused recurrent state, isolated valid graph nodes, and highly separable generated events [5]. HGF-ZDC was structurally stronger but retained a large response bias, late-HCAL suppression, loss–physics misalignment, weak speed, and an invalid fixed-condition reference construction [6]. The supplied single-stage graph-flow proposal remains a necessary controlled baseline because it tests whether explicit hierarchy earns its cost on the same data and geometry [7].

2 Physics basis: neutrons, hadronic showers, ZDCs, and Geant4

2.1 Why neutrons make the longitudinal problem stochastic

A high-energy neutron is electrically neutral and therefore does not continuously ionize like a charged particle before a hadronic interaction. In Geant4, a hadronic process combines cross sections that determine interaction occurrence with final-state models that generate the products; multiple models cover different energy ranges and physical regimes [4]. Consequently, two

neutrons with the same macroscopic four-momentum can have different first visible interaction depths, secondary multiplicities, electromagnetic fractions, nuclear breakup, invisible energy, transverse widths, and leakage.

The average longitudinal profile of a calorimeter shower can often be summarized by smooth parameterizations, including gamma-family forms, but event-level hadronic profiles are neither deterministic nor layerwise monotone [8, 9]. A neutron may deposit little energy in early material, initiate a cascade later, rise toward a shower maximum, and then decay. Therefore the constraint

$$D_{\ell+1} \leq D_{\ell} \quad \text{for all layers } \ell \quad (2)$$

is rejected. It would make a late-start event impossible: once an early D_{ℓ} is near zero, all subsequent deposits would be forced near zero.

2.2 Role of a Zero Degree Calorimeter

ZDCs are forward hadronic calorimeters designed to measure neutral or charged spectator nucleons close to the beam direction. Their response is detector-specific: geometry, segmentation, absorber, active material, acceptance, beam conditions, and readout all shape the observed spectrum and resolution [10, 11]. Published ALICE ZDC generative work is therefore an important methodological baseline but not a drop-in representation for a detector with 6,790 irregular readout channels [12, 13, 14, 15, 16].

2.3 Why the model should emulate the scored target, not microscopic transport

The surrogate target is the distribution of stored detector outputs, not an unsupervised reconstruction of Geant4’s internal particle history. Learned “transport,” “branching,” or “energy flow” should be interpreted as statistical dependence unless track-level or step-level truth supervises a microscopic claim. The model may incorporate detector depth, adjacency, and remaining budget as inductive biases, but it remains a conditional density estimator for the scored response.

3 Data contract and supported condition manifold

3.1 Sole raw condition

The public event interface is exactly

$$p^{\mu} = (E, p_x, p_y, p_z) \in \mathbb{R}^4. \quad (3)$$

For a neutron of mass m_n , preprocessing verifies the mass-shell relation

$$E^2 - p_x^2 - p_y^2 - p_z^2 = m_n^2, \quad (4)$$

within a tolerance justified by source precision. The minimal derived feature vector is

$$\xi(p^{\mu}) = \left(\log \frac{E}{E_0}, \frac{p_x}{\|\mathbf{p}\|}, \frac{p_y}{\|\mathbf{p}\|}, \frac{p_z}{\|\mathbf{p}\|} \right), \quad (5)$$

followed by an encoder $c = f_{\text{cond}}(\xi)$. Angles, slopes, β , γ , momentum magnitude, and a mass-shell residual may be computed for QA or ablation, but they are algebraically derived rather than independent inputs.

3.2 Static detector metadata

The detector definition $\mathcal{G}$ contains the ordered list of valid readout channels and their static features. The intended contract is

$$N_{\text{ECAL}} = 400, \quad N_{\text{HCAL}} = 6390, \quad N = 6790, \quad (6)$$

with 65 longitudinal layers in total. Per-node metadata should include, where available,

$$g_i = (x_i, y_i, z_i, \ell_i, \text{subdetector}_i, \text{area}_i, \text{ganging}_i, \text{valid}_i, \dots). \quad (7)$$

Graph edges must be generated from audited physical adjacency or overlap, not from array-slot coincidence. Lateral edges may be bidirectional. Longitudinal edges used in a causal field should point from shallower to deeper layers; a separate bidirectional graph may be tested as an ablation.

3.3 Entry position derived from four-momentum

If the neutron gun vertex (x_0, y_0, z_0) and detector plane $z = z_{\text{det}}$ are fixed, a straight-line intercept is a deterministic feature:

$$x_{\text{entry}} = x_0 + (z_{\text{det}} - z_0) \frac{p_x}{p_z}, \quad (8)$$

$$y_{\text{entry}} = y_0 + (z_{\text{det}} - z_0) \frac{p_y}{p_z}. \quad (9)$$

This does not create independent entry-position coverage. The claimed condition domain must be the manifold actually present in the Geant4 generator. Arbitrary edge-entry claims require new Geant4 data in which position and direction are varied with recorded provenance.

3.4 Target mode must be frozen before training

Two output contracts are scientifically distinct.

Raw-deposit mode. Y_i is the raw Geant4 energy deposit in readout channel i . Exact zeros are a point mass. The anti-dust threshold is $\tau = 0$; exact support is still required because an ordinary softmax is positive in every component.

Thresholded-readout mode. A detector, electronics, or analysis threshold $\tau > 0$ is frozen before training:

$$Y_i^{(\tau)} = Y_i \mathbf{1}\{Y_i \geq \tau\}. \quad (10)$$

For raw-energy accounting, the layer-level subthreshold remainder is

$$U_\ell = \sum_{i:\ell_i=\ell} Y_i \mathbf{1}\{0 < Y_i < \tau\}. \quad (11)$$

The threshold must not be chosen simply because it makes optimization easier.

3.5 Energy support assumption

The reference implementation uses a response fraction $\rho = T/E \in [0, 1]$. This support is enabled only if a full data audit shows that the stored target is raw deposited energy and that legitimate $T > E$ events do not occur, apart from a documented numerical tolerance. Nuclear Q values and the exact scoring contract make this an empirical data-contract question, not an axiom. If the audit reveals legitimate overflow, replace the bounded model with a target-correct distribution or an explicit overflow component; do not clip truth silently.

4 Statistical target and factorization

Let V be a visible-response indicator; S the first visible layer; $A = (A_0, \dots, A_{L-1})$ the active-layer support; T total modeled response; $D = (D_0, \dots, D_{L-1})$ layer budgets; R_{res} an unallocated or leakage-accounting reserve; $K = (K_0, \dots, K_{L-1})$ resolved hit counts; $\mathcal{S}_\ell$ the selected cell set in layer ℓ ; and Y the final cell vector. With condition $c = f_{\text{cond}}(p^\mu)$ and shared event latent $z \sim \mathcal{N}(0, I)$, the proposed factorization is

$$\begin{aligned} p_\theta(Y, V, S, A, T, D, R_{\text{res}}, K, \mathcal{S} | c, \mathcal{G}) \\ = p(V | c, z) p(T | V, c, z) p(S | T, V, c, z) p(A | S, T, V, c, z) p(D, R_{\text{res}} | A, T, c, z) \\ \times p(K | D, A, c, z) p(\mathcal{S} | K, D, A, c, z, \mathcal{G}) p(Y | \mathcal{S}, K, D, c, z, \mathcal{G}). \end{aligned} \quad (12)$$

The shared z is not required to be interpreted physically. Its purpose is to preserve event-level dependence among response, start depth, occupancy, and morphology rather than sampling all stages independently.

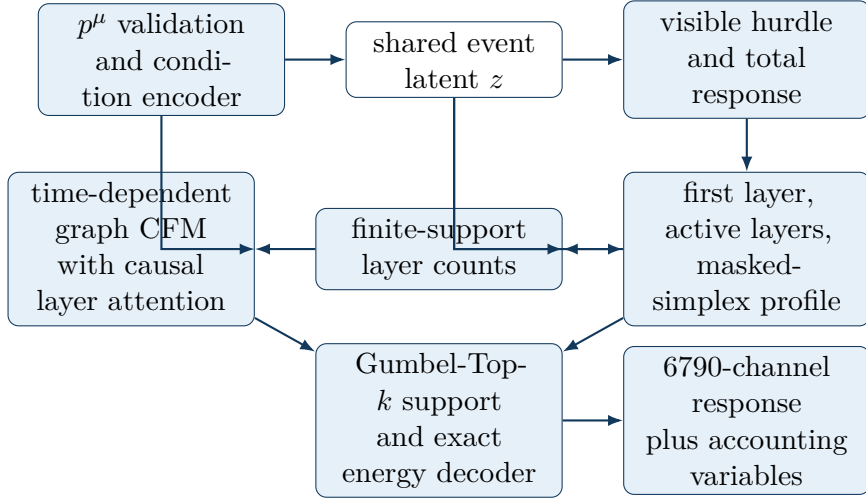


Figure 1: CBSC-ZDC factorization. All layer and cell outputs are generated conditionally on the same event condition and stochastic latent. The flow field updates all nodes in parallel at each solver step.

5 Longitudinal cascade model

5.1 Visible-response hurdle

The event-level response point mass is

$$V \sim \text{Bern}(\pi_V(c, z)), \quad \pi_V = \text{sigmoid}(a_V(c, z)). \quad (13)$$

When $V = 0$, the modeled cell response, layer budgets, and counts are all zero. A separate no-response component is preferable to asking a continuous density to approximate a delta function at zero.

5.2 Total response

For the audited bounded mode, let $\rho = T/E \in (0, 1)$. A flexible known choice is a finite mixture of beta distributions:

$$p(\rho | V = 1, c, z) = \sum_{m=1}^M \pi_m(c, z) \text{Beta}(\rho; \alpha_m(c, z), \beta_m(c, z)), \quad (14)$$

$$\pi = \text{softmax}(a_\pi), \quad \alpha_m = \text{softplus}(a_{\alpha,m}) + \alpha_{\min}, \quad (15)$$

$$\beta_m = \text{softplus}(a_{\beta,m}) + \beta_{\min}, \quad T = VE\rho. \quad (16)$$

If the data have an endpoint mass at $\rho = 1$, an explicit endpoint component should be added instead of relying on clipping.

5.3 First-visible-layer survival model

Let $h_\ell \in (0, 1)$ denote a discrete hazard:

$$h_\ell = p(S = \ell | S \geq \ell, V = 1, c, z, T). \quad (17)$$

Then the unnormalized probability of starting at layer ℓ inside the detector is

$$\tilde{p}_\ell = h_\ell \prod_{j < \ell} (1 - h_j), \quad (18)$$

and $p(S = \ell | S < L)$ is obtained by normalizing $\tilde{p}_\ell$ across detector layers. This variable is the first *visible* layer. It must not be described as the microscopic first nuclear interaction unless such truth is available.

5.4 Exact active-layer support

Conditional on S , the model generates $A_\ell \in \{0, 1\}$ with

$$A_\ell = 0 \quad (\ell < S), \quad A_S = 1. \quad (19)$$

For $\ell > S$, activity may be generated by correlated Bernoulli logits conditioned on (c, z, T, S) , a causal discrete model, or a compact conditional flow over relaxed support variables followed by discrete sampling. The baseline implementation uses shared-latent Bernoulli logits and enforces the start constraints exactly. Because gaps can occur in sparse readout, the model should not force every post-start layer active unless the audited target supports that assumption.

5.5 Masked-simplex energy allocation

Let $q_0, \dots, q_{L-1}, q_{\text{res}}$ be profile logits. Inactive layers receive $-\infty$ before normalization:

$$\bar{q}_\ell = \begin{cases} q_\ell, & A_\ell = 1, \\ -\infty, & A_\ell = 0. \end{cases} \quad (20)$$

Define

$$(w_0, \dots, w_{L-1}, w_{\text{res}}) = \text{softmax}(\bar{q}_0, \dots, \bar{q}_{L-1}, q_{\text{res}}), \quad (21)$$

then

$$D_\ell = Tw_\ell, \quad R_{\text{res}} = Tw_{\text{res}}. \quad (22)$$

A logistic-normal profile is the reference sampler. A low-dimensional conditional flow-matching profile over the same masked-simplex target is a planned comparison, motivated by hierarchical calorimeter generators such as CaloFlow, iCaloFlow, L2LFlows, and CaloDREAM [17, 18, 19, 20, 21].

Proposition 5.1 (Monotone remaining budget). *Assume $T \geq 0$, $D_\ell \geq 0$, and $\sum_{\ell=0}^{L-1} D_\ell + R_{\text{res}} = T$. Define*

$$R_\ell = T - \sum_{j=0}^{\ell} D_j. \quad (23)$$

Then $R_{\ell+1} \leq R_\ell$ for every ℓ , $R_\ell \geq R_{\text{res}} \geq 0$, and no ordering restriction is imposed on the individual D_ℓ .

Proof. Since $D_{\ell+1} \geq 0$, $R_{\ell+1} = R_\ell - D_{\ell+1} \leq R_\ell$. Moreover, $R_\ell = R_{\text{res}} + \sum_{j>\ell} D_j \geq R_{\text{res}} \geq 0$. The result contains no inequality comparing $D_{\ell+1}$ with D_ℓ . $\square$

Remark 5.2. The reserve is an accounting variable. Its scientific interpretation must be frozen: it may represent unmodeled/leakage response, subdetector-external energy, or simply a profile slack variable. It must not be called physical leakage without matching truth.

6 Finite-support count model

For each layer ℓ with N_ℓ valid channels, generate

$$K_\ell \sim \text{Cat}(\eta_{\ell,0}, \dots, \eta_{\ell,N_\ell}), \quad (24)$$

where logits depend on $(c, z, D_\ell, A_\ell, \ell)$. A categorical count avoids the Poisson restriction $\text{Var}(K) = \mathbb{E}[K]$ and permits exact masking of impossible values.

For thresholded mode, the resolved-count contract is piecewise:

$$K_\ell = \begin{cases} 0, & 0 \leq D_\ell < \tau, \\ 1, \dots, \min(N_\ell, \lfloor D_\ell/\tau \rfloor), & D_\ell \geq \tau. \end{cases} \quad (25)$$

The first branch routes the layer budget to the recorded subthreshold residual rather than inventing a resolved cell. For raw-deposit mode, $K_\ell = 0$ iff $D_\ell = 0$, while $1 \leq K_\ell \leq N_\ell$ whenever $D_\ell > 0$. The count distribution is a central physics object, not an implementation detail, because exact support decoding transfers occupancy fidelity into the count model.

7 Parallel conditional flow-matching spatial model

7.1 Why parallel rather than a 65-call rollout

Layer-to-layer autoregression is a known and sometimes effective strategy [18, 19, 20]. For this detector it carries three risks: invalid index correspondence across irregular layers, accumulated free-running error, and serial latency. The proposed field updates all nodes at every numerical integration step, while causal layer attention and directed longitudinal graph edges preserve the intended dependence on earlier depth.

7.2 Continuous target state

The flow state has two channels per node:

$$X_i = (s_i, r_i), \quad (26)$$

where s_i is a continuous support-ranking variable and r_i is a positive-energy share variable. One supervised encoding is

$$s_i^* = \text{logit}(\varepsilon + (1 - 2\varepsilon)M_i), \quad (27)$$

$$r_i^* = M_i \left[\log(Y_i - \tau + \varepsilon) - \frac{1}{K_{\ell_i}} \sum_{j \in \mathcal{S}_{\ell_i}} \log(Y_j - \tau + \varepsilon) \right], \quad (28)$$

with $M_i = \mathbf{1}\{i \in \mathcal{S}_{\ell_i}\}$. Equation (28) is evaluated only for layers with $K_{\ell_i} > 0$; on zero-count layers set $r_i^* = 0$ and omit the positive-share term from the loss. Other monotone encodings may be tested, but the decoder and target transform must be versioned together.

7.3 Flow-matching objective

Conditional flow matching learns a time-dependent velocity field without evaluating a simulation likelihood along an ODE path [22, 23]. For a straight interpolation,

$$X_0 \sim \mathcal{N}(0, I), \quad t \sim \mathcal{U}(0, 1), \quad (29)$$

$$X_t = (1 - t)X_0 + tX_1, \quad (30)$$

$$u_t = X_1 - X_0. \quad (31)$$

The vector field is

$$v_\theta = v_\theta(X_t, t, c, z, \mathcal{G}, D, K), \quad (32)$$

and the core loss is

$$\mathcal{L}_{\text{FM}} = \mathbb{E} \left[\|v_\theta(X_t, t, c, z, \mathcal{G}, D, K) - u_t\|_W^2 \right], \quad (33)$$

where W may balance support and positive-share channels. The network must explicitly consume t ; an implementation that accepts but ignores t is not the specified field.

7.4 Velocity network

At each solver step:

1. concatenate X_t with static node features and layer-level dynamic features $\log(1 + D_\ell)$ and K_ℓ/N_ℓ ;
2. add a learned time embedding and apply FiLM conditioning from c and, if separate, z ;
3. apply edge-conditioned graph message passing using audited lateral and directed longitudinal edges;
4. pool node states into layer tokens;
5. apply a transformer with a causal layer mask, so layer ℓ attends only to layers $0, \dots, \ell$;
6. broadcast the mixed layer token to its nodes and predict the two velocity channels.

This combines established graph, transformer, and conditional-flow components rather than proposing a new primitive [24, 21, 25].

7.5 Sampling

Starting from $X(0) \sim \mathcal{N}(0, I)$, solve

$$\frac{dX}{dt} = v_\theta(X, t, c, z, \mathcal{G}, D, K), \quad t \in [0, 1]. \quad (34)$$

Euler, midpoint, Heun, or an adaptive solver may be used. The solver, step count, precision, and random seed must be included in every timing and fidelity report. Published ZDC flow matching demonstrates that low-step and latent variants can achieve large speed gains on compact ALICE responses, but those numbers are not directly transferable to a 6,790-node detector [14, 26].

8 Exact sparsity and the anti-dust decoder

8.1 The failure mode

A dense softmax allocation gives every valid cell positive energy. A model can then improve an aggregate energy loss by assigning a small value, such as 10^{-3} GeV, to many nominally empty cells. This can leave the total approximately correct while corrupting occupancy, positive-hit spectra, width, connectedness, and downstream reconstruction. Sparse HEP models explicitly show the value of modeling the point mass at zero [27, 28, 29]. Calorimeter-flow studies also report that post-generation rescaling can lift nominally zero voxels and create excess low-energy hits [18, 20].

8.2 Primary exact-support construction

For layer ℓ , let $B = D_\ell$ be the layer budget, $K = K_\ell$ the feasible count, $\mathcal{I}_\ell$ the valid channel set, and a_i support logits. Select a subset $\mathcal{S}_\ell \subset \mathcal{I}_\ell$ with $|\mathcal{S}_\ell| = K$ by Gumbel-Top- k sampling without replacement [30]:

$$\mathcal{S}_\ell = \text{TopK}_{i \in \mathcal{I}_\ell}(a_i + g_i), \quad g_i \sim \text{Gumbel}(0, 1). \quad (35)$$

For selected-cell share logits r_i , define

$$w_i = \frac{e^{r_i}}{\sum_{j \in \mathcal{S}_\ell} e^{r_j}}, \quad i \in \mathcal{S}_\ell. \quad (36)$$

Decode

$$\hat{Y}_i = \begin{cases} \tau + (B - K\tau)w_i, & i \in \mathcal{S}_\ell, \\ 0, & i \notin \mathcal{S}_\ell, \end{cases} \quad (37)$$

provided $B \geq K\tau$. If $B < \tau$, the feasible count is zero, every cell is zero, and B is recorded as the layer-level subthreshold residual U_ℓ .

Theorem 8.1 (No-dust, exact-count, exact-budget decoder). *Consider a resolved layer with $B > 0$ and an integer K satisfying $1 \leq K \leq N_\ell$ and $B \geq K\tau$. Let $\mathcal{S}_\ell$ have cardinality K , and let $w_i > 0$ on $\mathcal{S}_\ell$ with $\sum_{i \in \mathcal{S}_\ell} w_i = 1$. Then the decoder in equation (37) satisfies:*

- (i) $\hat{Y}_i = 0$ for $i \notin \mathcal{S}_\ell$;
- (ii) $\hat{Y}_i \geq \tau$ for $i \in \mathcal{S}_\ell$;
- (iii) $\sum_{i \in \mathcal{I}_\ell} \hat{Y}_i = B$;
- (iv) the number of positive cells is exactly K ;
- (v) no cell satisfies $0 < \hat{Y}_i < \tau$.

For an unresolved thresholded layer with $0 \leq B < \tau$, define $K = 0$, $\hat{Y}_i = 0$ for every cell, and $U_\ell = B$. For the zero-budget case in either target mode, define $K = 0$, $\hat{Y}_i = 0$, and $U_\ell = 0$.

Proof. Items (i) and (ii) follow directly from the piecewise definition and $B - K\tau \geq 0$. For (iii),

$$\sum_i \hat{Y}_i = K\tau + (B - K\tau) \sum_{i \in \mathcal{S}_\ell} w_i = B. \quad (38)$$

Because $K \geq 1$ and every selected weight is positive, exactly the K selected cells are positive. Item (v) follows because each cell is either zero or at least τ . The residual case is the stated definition. $\square$

Corollary 8.2 (Event accounting). *If each layer is decoded as above, then*

$$\sum_{i=1}^N \hat{Y}_i + \sum_{\ell=0}^{L-1} U_\ell + R_{\text{res}} = T. \quad (39)$$

8.3 What exact Top- k does not guarantee

The decoder guarantees exact support *conditional on the generated count*. It cannot prevent a poorly trained count network from requesting too many cells. In particular, a model could choose $K_\ell = N_\ell$ and remain algebraically valid while producing physically incorrect occupancy. The experiment therefore trains and reports the count as an independent random variable. For a frozen target threshold,

$$K_\ell^* = \sum_{i:\ell_i=\ell} \mathbf{1}\{Y_i \geq \tau\} \quad (40)$$

(with strict positivity replacing $Y_i \geq \tau$ when $\tau = 0$), and $p_\theta(K_\ell \mid c, z, D_\ell, A_\ell)$ is optimized by categorical negative log-likelihood. Requested-count calibration, realized-count calibration, and occupancy distributions are reported separately from the total loss. Support logits are supervised or ranked against the corresponding truth set. Later training exposes the spatial field to generated counts and generated support so that good truth-conditioned behavior cannot conceal free-running support errors. Thus Top- k removes dense low-energy dust, while the count and support objectives prevent the model from evading that constraint by declaring every cell active.

8.4 Training the discrete support

Hard Top- k is non-differentiable through the selected indices. It is therefore the exact sampling and evaluation decoder, not by itself a complete training objective. Recommended known options are:

1. supervise support logits with class-balanced binary cross entropy or ranking loss while training positive shares on truth support;
2. use a continuous subset relaxation or straight-through estimator during joint fine-tuning [31];
3. expose the positive-share model to generated counts and generated support during later stages to reduce truth-mask/sample-mask mismatch;
4. compare exact Top- k with sparsemax/entmax and hard-concrete gates [32, 33, 34].

The primary specification retains Top- k because its support size, zeros, and accounting are directly auditable. The alternatives are ablations, not assumed upgrades.

9 Training objectives

The complete objective is modular:

$$\begin{aligned} \mathcal{L} = & \lambda_V \mathcal{L}_V + \lambda_\rho \mathcal{L}_\rho + \lambda_S \mathcal{L}_S + \lambda_A \mathcal{L}_A + \lambda_D \mathcal{L}_D + \lambda_K \mathcal{L}_K \\ & + \lambda_{\text{FM}} \mathcal{L}_{\text{FM}} + \lambda_{\text{sup}} \mathcal{L}_{\text{sup}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}. \end{aligned} \quad (41)$$

The terms are:

$$\mathcal{L}_V = \text{BCE}(V, \pi_V), \quad (42)$$

$$\mathcal{L}_\rho = -\log p(\rho \mid V = 1, c, z), \quad (43)$$

$$\mathcal{L}_S = -\log p(S \mid V = 1, c, z, T), \quad (44)$$

$$\mathcal{L}_A = \text{BCE}(A, \pi_A) \text{ on allowed layers}, \quad (45)$$

$$\mathcal{L}_K = \sum_{\ell} \text{CE}(K_\ell, \eta_\ell), \quad (46)$$

$$\mathcal{L}_{\text{sup}} = \text{BCE}(M, \sigma(s)) \text{ or a ranking/relaxed-set loss}. \quad (47)$$

$\mathcal{L}_D$ may be a masked-simplex likelihood, profile flow-matching loss, or a set of conditional distributional terms. $\mathcal{L}_{\text{aux}}$ may include moment or positive-hit-spectrum diagnostics, but it must not reduce the experiment to eventwise regression against an arbitrary independent Geant4 shower.

All unweighted component losses, weighted contributions, and stage-wise gradient norms should be reported. Loss weights should be selected on a frozen validation protocol, not tuned against the final test set. A lower scalar loss is not sufficient evidence of improved FastMC fidelity; the previous HGF-ZDC run supplied a direct counterexample [6].

10 Data pipeline and provenance

10.1 Raw ROOT audit

Before model training, the converter should report:

1. file hashes, ROOT tree name, branch names, types, units, and entry counts;
2. four-momentum mass-shell residuals and energy/angular support;
3. exact channel-ID decoding, duplicate IDs, invalid IDs, sentinel records, and energy carried by excluded records;
4. agreement between reconstructed ECAL/HCAL totals and producer summary branches;
5. zero-response frequency, total-response support, layer occupancy, and threshold sensitivity;
6. fixed versus variable gun vertex and the implied condition manifold;
7. available job, run, event, and random-seed provenance for leakage-safe splitting.

The small attached ROOT file has the same branch structure but a different Geant4 run and must be used only as a schema/reference fixture. In the current execution environment, ROOT/Uproot/Awkward were not available for a fresh numerical decode; the repository therefore contains a branch-map-oriented adapter scaffold rather than fabricated event statistics.

10.2 Canonical tensors

A canonical event contains at least:

p4	float32	[4]	
cell_energy_raw	float32	[6790]	
cell_energy_tau	float32	[6790]	# only in thresholded mode
layer_energy	float32	[65]	
layer_hit_count	int64	[65]	
active_layer	bool	[65]	
first_visible	int64		
subthreshold	float32	[65]	# thresholded mode
metadata	event/run/source	provenance	

Channel order and geometry version must be immutable and hashed. Padding must not be confused with valid zero-energy cells.

10.3 Splitting

The preferred split unit is an independent Geant4 generation group, such as job/run/seed family. If no grouping metadata exist, a deterministic row-index hash split may be unavoidable, but its limitation must be reported and duplicate/correlation searches should be performed. All data-derived normalization, thresholds, mixture initialization, and model selection use training and validation data only. The final test manifest is read only for the final declared evaluation.

10.4 Training range experiment

Run two matched experiments:

R1. train on 50–250 GeV only;

R2. train on the full supported 0–300 GeV range while retaining the same architecture and primary reporting domain.

Broader support may reduce boundary artifacts but may also consume capacity on qualitatively different low-energy response. The effect must be measured rather than assumed. This has no direct relationship to collider event pileup: training on a wider single-neutron energy range does not itself teach superposition of multiple particles. A pileup-capable model would require multi-particle conditioning or a validated composition procedure.

11 Staged implementation and Vertex AI execution plan

11.1 Stage sequence

The recommended sequence is:

Stage	Work	Recommended report
0	Freeze target, threshold, geometry, split, and provenance.	Data-contract report with hashes, branch map, support plots, sentinel accounting, and condition-manifold statement.
1	Train visible hurdle and total-response model.	Conditional zero rate, response quantiles, means, variances, tails, calibration, and train/validation uncertainty.
2	Train first-visible, active-layer, and masked-simplex profile.	Start-depth distribution, per-layer zero rate, energy profiles, cumulative profiles, late energy, and covariance.
3	Train finite-support count model.	Count calibration and distributions by energy, angle, layer, start depth, and layer budget.
4	Train spatial CFM using truth profile and truth counts.	Time-dependent loss, support ranking, positive-hit spectrum, geometry-aware morphology, memory, and throughput.
5	Expose spatial model to generated counts/support and generated profile.	Difference between truth-conditioned and free-running results; support mismatch and error propagation.
6	Optional joint fine-tuning.	All component losses and gradients plus full-cascade free-running metrics.
7	Baselines and ablations.	Matched data, optimization steps, hardware, precision, solver, parameters, memory, and wall time.
8	Final held-out and downstream study.	Primary-domain, stress-domain, reconstruction, uncertainty, and timing reports kept separate.

These are reporting locations, not user-imposed pass/fail gates. Algebraic assertions such as finite tensors and exact accounting remain software invariants because violating them means the implementation no longer matches the specification.

11.2 Vertex job layout

Each Vertex job should be immutable and self-describing:

```
runs/<experiment_id>/
  config_resolved.yaml
  environment.txt
  git_commit.txt
  data_manifest.json
  geometry_hash.txt
  statistics.pt
  checkpoints/
  histories/
  free_running_validation/
  timing/
  logs/
```

Use checkpoint resume with optimizer, scheduler, scaler, random-number-generator states, epoch/step counters, and validation-bank version. Periodic checkpoint upload should write only changed artifacts rather than resynchronizing the complete output tree.

11.3 Precision and performance

Begin with float32 accumulation in graph reductions and profile/accounting operations. Mixed precision is introduced only after finite-gradient and output-equivalence probes, because earlier project work found several graph/autocast dtype and accumulation failures [6]. Report model-only and end-to-end timing separately at batch 1 and representative larger batches.

12 Baselines and controlled comparisons

12.1 Direct ZDC references

The following repositories are the highest-priority external comparisons:

1. `m-wojnar/faster_zdc`: full and latent flow-matching models accompanying the ALICE ZN/ZP study [14, 26];
2. `m-wojnar/zdc`: VAE, GAN, vector-quantized, and diffusion-era ALICE ZDC work [35, 12, 13];
3. ExpertSim’s mixture-of-generative-experts implementation for heterogeneous ZDC responses [15];
4. the inverse-autoregressive-flow ZDC line, where code availability and license must be confirmed before integration [16].

These should first be reproduced on their native representations. Any adaptation to 6,790 irregular channels is a new experiment and must not be called a direct reproduction.

12.2 Internal matched baseline ladder

On the identical training/validation manifests, optimization steps, solver steps, precision, and hardware, compare:

B0: conditional empirical/template baseline

A non-neural or low-capacity model for visible probability, total response, profile, count, and an entry-centered spatial template with empirical residuals.

B1: non-graph parallel CFM

Same profile, count, state encoding, and exact decoder as CBSC-ZDC; remove graph message passing and retain only node features/layer tokens.

M1: mixture-of-experts

Route broad response regimes to specialized versions of a compact known generator, motivated by ExpertSim [15].

G1: CBSC-ZDC

Full graph plus causal layer attention.

S1: single-stage graph flow

The supplied baseline that removes the start/profile hierarchy and learns the full response in one graph flow [7].

A1: serial previous-layer model

A diagnostic layer-autoregressive ablation using valid geometric correspondence; included to measure fidelity/latency, not assumed superior.

P1: sparse point-cloud model

Optional comparison if the hit threshold and readout mapping are frozen, motivated by CaloClouds and CaloPointFlow [29, 36, 28].

The graph is justified only if it improves predeclared geometry-sensitive results enough to compensate for its latency and memory. Production fast simulation is often hybrid rather than architecturally pure; AtlFast3 is an important precedent [37].

13 Recommended QA and reporting

13.1 Software and algebraic QA

At construction and sampling, report or assert:

- tensor shapes, dtypes, devices, masks, and edge ranges;
- no NaN or infinity and no negative cell energy;
- zero energy in invalid or padded channels;
- exact support-count equality;
- zero forbidden dust cells $0 < Y_i < \tau$ in thresholded mode;
- per-layer equality between resolved cell energy and $D_\ell - U_\ell$;
- event identity in [equation \(39\)](#);
- $T \leq E$ only when the bounded raw-deposit assumption has been audited;
- deterministic replay under fixed seeds and diversity under different seeds;
- explicit dependence of the velocity field on flow time t .

13.2 Training-time monitoring

A frozen validation bank should be generated in free-running mode during training. Report:

1. all component losses and stage-wise gradient norms;
2. total-response mean, width, quantiles, and zero rate by condition;
3. first-visible layer, active-layer count, longitudinal profile, and late-energy fraction;
4. hit count, positive-hit spectrum, cell occupancy, and subthreshold residual;
5. transverse centroid, RMS, R_{50} , R_{90} , maximum-cell and top- k fractions;
6. correlations among response, start depth, hit count, width, ECAL fraction, and late energy;
7. inference time, memory, and numerical failures.

A checkpoint is not described as improving merely because the training loss decreases. Report paired/bootstrap uncertainty for differences between checkpoints and label the result as improvement, plateau, regression, or inconclusive according to the frozen analysis plan.

13.3 Distributional evaluation

Use complementary diagnostics rather than one aggregate distance:

- one-dimensional conditional histograms and distances for named physics observables;
- high-level classifier two-sample tests;
- geometry-aware low-level graph, sparse, or point-cloud classifiers;
- Fréchet and kernel physics distances with a frozen disclosed feature extractor [38];
- precision/recall or coverage diagnostics to distinguish realistic but incomplete generation from broad unphysical generation [39, 40];
- nearest-neighbor train/test comparisons for memorization;
- truth-half versus truth-half statistical floors;
- repeated-condition Geant4 and FastMC ensembles for conditional variance and tails.

Averaging an independent per-cell distance over thousands of mostly zero channels is not sufficient evidence of joint spatial closure.

13.4 Downstream and amplification studies

Freeze the reconstruction model before comparing Geant4 and FastMC. Report reconstructed energy scale/resolution, position/angle resolution, efficiencies, threshold behavior, and tail migration. Also perform train-on-Geant4/test-on-FastMC and train-on-FastMC/test-on-locked-Geant4 studies. Recent work emphasizes that full-physics reconstruction benchmarks can reveal failures hidden by shower-image metrics [41].

Generating many more FastMC events than training events reduces sampling noise but not surrogate bias. Any “calomplification” claim must be observable-specific and measured against an independent larger Geant4 reference [42].

14 Failure modes and required interpretations

Failure mode	Symptom	Required diagnosis or mitigation
Dense energy dust	Many cells contain tiny positive values; occupancy and width inflate.	Exact count/support decoder; threshold contract; positive-hit and subthreshold reports.
Unbounded response tail	Rare samples dominate the mean or exceed physically audited support.	Target-correct response distribution; profile-only million-sample tail audit.
Profile underweighting	Global flow loss improves while response bias stays fixed.	Separate response objective/gradients; standalone response model; truth-profile cross-tests.
Mode averaging	Correct pooled mean, wrong zero/late/leakage regimes.	Hurdle and mixture routing; conditional precision/recall; repeated conditions.
Count bottleneck	Exact decoder produces wrong morphology because K_ℓ is wrong.	Count calibration; truth-count versus generated-count spatial tests; compare sparse alternatives.
Exposure bias	Truth-conditioned stage is good, full cascade degrades.	Generated-condition exposure, cross-combination matrix, optional joint fine-tuning.
Invalid graph	Isolated valid nodes or readout-index edges.	Geometry audit, degree/connectivity report, physical edge provenance, no-graph comparison.
Condition ignorance	Pooled samples look plausible but anchors do not change with p^μ .	Condition permutation, local interpolation, repeated-p4 evaluation.
Support extrapolation	Edge or angle claims outside Geant4 coverage.	Report as stress/OOD; generate new anchored Geant4 data.
Latency failure	Graph/ODE model is slower than useful alternatives.	Profile every stage; solver sweep; distillation/latent comparison only after fidelity teacher exists.

15 Implementation contract

15.1 Reference tensor interface

Object	Shape	Contract
p^μ	$[B, 4]$	Raw condition in GeV; neutron mass-shell validated.
Condition embedding c	$[B, C]$	Derived only from p^μ .
Node features	$[6790, F_n]$	Static geometry; versioned and hashed.
Edge index	$[2, N_e]$	Static valid node IDs; edge direction/-type documented.
Edge features	$[N_e, F_e]$	Relative displacement, distance, overlap/type encodings.
Layer index	$[6790]$	Integer in $[0, 64]$.
Profile budgets D	$[B, 65]$	Nonnegative; zero on inactive layers.

Object	Shape	Contract
Counts K	$[B, 65]$	Integer and feasible by geometry/threshold.
Flow state X_t	$[B, 6790, 2]$	Support-ranking and share channels.
Final cell energy Y	$[B, 6790]$	Nonnegative, exact zeros outside support.
Residual U	$[B, 65]$	Subthreshold energy in thresholded mode.
Reserve	$[B, 1]$	Profile accounting channel with frozen interpretation.

15.2 Reference hyperparameters for the first pilot

These are starting values for a matched pilot, not empirically established optima:

Component	Initial specification
Condition embedding	$C = 128$, residual MLP, SiLU, LayerNorm
Shared latent	32 Gaussian dimensions
Response mixture	4 beta components plus visible hurdle
Profile network	hidden 256; logistic-normal or low-dimensional CFM
Count network	hidden 192; categorical support $0, \dots, N_\ell$
Spatial hidden width	96
Graph blocks	2 edge-conditioned residual blocks
Layer transformer	3 blocks, 4 heads, causal mask
Flow solver	compare 4, 8, and 16 evaluations
Primary state	two channels per node
Support sampling	stochastic Gumbel-Top- k ; deterministic Top- k for diagnostics
Precision	float32 reference; mixed precision only after equivalence tests
Pilot data	matched 50k–100k-event studies

15.3 Pseudocode

```

function SAMPLE(p4, geometry, seed):
    validate_units_shape_mass_shell(p4)
    c = condition_encoder(minimal_features(p4))
    z = Normal(0, I).sample(seed)

    V = sample_visible(c, z)
    rho = sample_target_correct_response(c, z, V)
    T = incident_energy(p4) * rho

    S = sample_first_visible_layer(c, z, T, V)
    A = sample_active_layers(c, z, T, S, V)
    D, reserve = sample_masked_simplex_profile(c, z, T, A)
    K = sample_feasible_layer_counts(c, z, D, A, threshold)

    X = Normal(0, I).sample(seed)
    for solver_time in configured_grid:
        velocity = spatial_graph_field(
            X, solver_time, c, geometry, D, K
        )
        X = solver_step(X, velocity)

```

```

support = gumbel_topk(X.support_scores, K, geometry.valid)
Y, residual = exact_threshold_safe_decode(
    support, X.share_scores, D, K, threshold
)
assert_structural_invariants(Y, residual, reserve, T)
return Y, diagnostics

```

16 Software QA status of the supplied scaffold

The revised repository is an executable reference scaffold, not a full Vertex training system. Static compilation and unit tests currently exercise:

- bounded response and exact profile accounting;
- first-visible-layer/start-support consistency;
- vectorized exact- K support decoding;
- no forbidden threshold dust and correct subthreshold residuals;
- valid-mask handling;
- time dependence of the flow field;
- edge-conditioned message passing and invalid-edge rejection;
- end-to-end sampling identities on synthetic geometry;
- fixed-seed replay.

The latest local command result is **30 passed**; measured source-and-test statement/branch coverage is 100% under the synthetic unit suite. PyTorch reports a nested-tensor optimization warning caused by the transformer’s `norm_first=True`. The warning does not indicate a numerical failure, but the production timing study should profile alternative transformer settings. Coverage demonstrates execution of the current scaffold paths, not fidelity to Geant4.

The hard decoder is vectorized by layer but remains an inference/evaluation operation. A production training entry point, ROOT branch map, canonical converter, distributed data loader, geometry builder, complete loss orchestration, Vertex launcher, and full evaluation package remain implementation work. This distinction is stated to prevent a tested scaffold from being misrepresented as a trained simulator.

17 Research literature audit and evidence depth

A prior project catalogue contains 600 linked works and a separate survey contains more than 100 named model instances. They are retained as broad discovery and contribution maps. They are not represented here as 600 line-by-line full-text peer reviews. The present specification relies on a smaller core bibliography whose claims were checked against primary papers, official documentation, or project evidence. The supplementary catalogue marks evidence depth, and auditors should not treat title/abstract or bibliography screening as equivalent to full-paper replication.

The broad survey nevertheless supports several robust synthesis points: there is no universal winner across CaloChallenge datasets and metrics; hierarchical global-to-local designs recur across successful flows and flow-matching models; diffusion and score models often offer high fidelity but require sampling acceleration; GANs are fast but vulnerable to coverage failure;

point-cloud and sparse models directly address empty-cell inefficiency; mixtures are relevant to heterogeneous ZDC responses; and production fast simulation is commonly hybrid [43, 44, 21, 25, 45, 36, 15, 37].

18 Five-minute explanation for a general science-fair judge

Detailed particle simulations are accurate but expensive because they track many random interactions through matter. This project asks whether a neural network can learn the distribution of what a neutron leaves in a forward calorimeter without retracing every microscopic step.

The detector is divided into thousands of readout cells and many depth layers. The model receives only the neutron’s energy and direction through its four-momentum. It first decides whether a visible shower occurs, how much total energy appears, where the shower begins, how that energy is shared across depth, and how many cells are active. A final generative model draws the detailed pattern.

The main physics correction is that energy in successive layers does not always decrease. A neutron can travel some distance before colliding, so the shower can start late and grow. What must decrease is the amount of energy still available to allocate. The model enforces that accounting exactly.

The model also prevents a common machine-learning shortcut. A network might put a tiny amount of energy into every cell to reduce its average error. Here it must first choose an exact number of hit cells; all other cells are exactly zero. The selected cells share the layer’s energy exactly. This makes the output auditable.

The test is not whether the images look convincing. The model will be compared with held-out Geant4 simulations on total response, shower depth, width, occupancy, rare tails, correlations, downstream reconstruction, and speed. Several published ZDC and calorimeter models will be run as baselines. The result may show that the proposed hierarchy helps, or that a simpler flow is better. Either outcome is scientifically useful because the comparison is controlled and falsifiable.

19 Five-minute explanation for a computational HEP physicist

The target is a conditional readout-level density $p_{G4}(Y \mid p^\mu, \mathcal{G})$ for single neutrons, with 6,790 nonnegative channels across one ECAL and 64 HCAL layers. The only raw event condition is p^μ ; geometry is static. The proposal is a known-component hierarchy designed around previously observed structural failures.

A visible-response hurdle precedes a target-audited total-response density, initially a beta mixture for $\rho = T/E$. A discrete survival model generates first visible depth. A correlated layer mask creates exact zero layers, and a masked simplex allocates T over active layers plus an accounting reserve. This makes the residual budget monotone without imposing the unphysical constraint $D_{\ell+1} \leq D_\ell$.

Counts are categorical with exact geometry- and threshold-feasibility masks. The spatial generator is a time-dependent CFM field on two node channels, conditioned on profile and counts. Each solver step applies edge-conditioned message passing and a causal layer-token transformer, updating all nodes in parallel. This replaces a serial 65-layer rollout while retaining longitudinal context.

At sampling, Gumbel-Top- k selects exactly K_ℓ cells. For a frozen threshold τ , selected energies are $\tau + (D_\ell - K_\ell\tau) \text{softmax}(r)$; unselected cells are zero. If $D_\ell < \tau$, the budget becomes a layer residual rather than voxel dust. The decoder proves exact count, threshold consistency, and energy closure. Training uses supervised/relaxed support objectives because hard Top- k is not differentiable.

The experiment separates profile-only, truth-profile spatial, generated-profile spatial, and full-cascade evaluations. Required comparisons are a competent empirical baseline, a matched

non-graph CFM, a response-regime mixture, the full graph model, the supplied single-stage graph flow, a valid serial ablation, and direct reproduction of published ALICE ZDC repositories on native data. No architecture is promoted by loss alone; free-running conditional distributions, geometry-aware C2ST, physics distances, coverage, reconstruction closure, and matched timing are reported. The specification guarantees structural validity under the target assumptions, not Geant4 fidelity.

20 Conclusions

The proposed experiment is worth implementing. Its defensible content is not a new neural primitive but a detector-specific synthesis of established ideas with exact contracts. The original monotone layer-deposit concept is replaced by monotone residual accounting. Exact layer support and finite counts prevent dense positive profiles. A threshold-safe Top- k decoder prevents artificial low-energy dust. Parallel graph-conditioned flow matching preserves longitudinal dependence without a serial rollout. The only raw condition remains four-momentum.

If implemented exactly, with a verified ROOT/geometry/target contract and the stated baseline comparisons, the project should yield a valid academic answer to whether this combination improves the speed-fidelity frontier for neutron ZDC simulation. A negative result would remain informative: it could show that the hierarchy, graph, exact count, or wider training range does not earn its complexity. What cannot be promised in advance is numerical agreement with Geant4. That must be established by the experiment.

A Claim-to-evidence map

Claim	Evidence or status
Geant4 hadronic interactions are stochastic and model/cross-section dependent.	Geant4 toolkit papers and official hadronic physics manual [1, 3, 4].
Neutron hadronic layer deposits need not decrease monotonically.	General hadronic-shower physics and longitudinal parameterization literature [8]; follows also from stochastic first interaction and cascade development.
ZDC response is detector-specific.	ALICE TDR and multinucleon/ZDC performance study [10, 11].
Hierarchical global/layer/local generation is established.	CaloFlow, iCaloFlow, L2LFlows, CaloDREAM [17, 18, 19, 21].
Conditional flow matching is an established training framework.	Flow Matching and minibatch OT-CFM [22, 23].
Sparse exact-support modeling addresses empty-cell structure.	SARM, CaloPointFlow, CaloClouds [27, 28, 29].
Gumbel-Top- k samples a fixed-size subset without replacement.	Kool et al. [30].
Sparsemax, entmax, hard-concrete, and relaxed subset methods are known alternatives.	[32, 33, 34, 31].
Published ALICE ZDC FM is a direct methodological baseline.	Wojnar paper and repository [14, 26].
Mixtures are relevant to heterogeneous ZDC response.	ExpertSim [15].

Claim	Evidence or status
Prior project models had documented structural and evaluation failures.	Internal independent audits [5, 6].
Exact decoder invariants hold.	Proved in this document and covered by synthetic unit tests; no external empirical claim.
CBSC-ZDC reproduces Geant4.	Not claimed; requires training and held-out evaluation.

B Chronological design corrections

1. Rejected monotone per-layer deposits after reviewing neutron/hadronic shower physics.
2. Retained monotone remaining-budget accounting, then replaced long multiplicative stick breaking with exact active support and masked-simplex allocation to avoid depth-order pressure.
3. Separated no-response, total response, first-visible depth, active layers, counts, and spatial allocation.
4. Corrected the spatial scaffold so the velocity field explicitly depends on flow time.
5. Added valid-node masking and edge-conditioned graph message passing with range checks.
6. Replaced nested event/layer support decoding with a layer-vectorized exact- K implementation.
7. Added first-visible-layer survival semantics and enforced zero activity before the sampled start.
8. Added the anti-dust threshold/residual contract and proved its invariants.
9. Clarified that hard Top- k is an inference/evaluation decoder and needs supervised or relaxed support training.
10. Separated structural correctness from empirical fidelity and removed any suggestion that unit tests validate the simulator.
11. Reclassified the 600-work catalogue as a discovery/contribution map rather than 600 full-text replications.

The repository’s chronological audit log records these externalized research and engineering decisions without purporting to reveal private internal reasoning.

C Recommended auditor checklist

1. Confirm the ROOT branch map, units, channel codec, sentinel meaning, and event counts.
2. Confirm that the target is raw deposit or thresholded readout, not a mixture of conventions.
3. Confirm the energy-support assumption before enabling $T/E \leq 1$.
4. Confirm geometry hashes and that every valid node has appropriate adjacency or is intentionally isolated.

5. Confirm the sole raw condition is p^μ and that any entry coordinate is deterministic under a fixed vertex.
6. Re-run static compilation, unit tests, and deterministic replay.
7. Trace every generated tensor to the mathematical variable defined in this paper.
8. Confirm that the flow field consumes t and that the solver is documented.
9. Confirm that hard support is not the only gradient path during training.
10. Confirm zero dust, exact count, per-layer accounting, and event accounting on a large generated sample.
11. Confirm primary-domain and stress-domain reports are separate.
12. Confirm the single-stage, no-graph, mixture, empirical, and external ZDC comparisons are matched or their mismatches disclosed.
13. Confirm the final test set was not used for preprocessing, threshold choice, or checkpoint selection.
14. Confirm every speed comparison reports hardware, batch, precision, steps, preprocessing, and output dimensionality.
15. Confirm conclusions do not exceed statistical precision or the supported condition manifold.

References

- [1] S. Agostinelli et al. “Geant4—a simulation toolkit”. In: *Nuclear Instruments and Methods in Physics Research A* 506 (2003), pp. 250–303. DOI: [10.1016/S0168-9002\(03\)01368-8](https://doi.org/10.1016/S0168-9002(03)01368-8).
- [2] J. Allison et al. “Geant4 developments and applications”. In: *IEEE Transactions on Nuclear Science* 53.1 (2006), pp. 270–278. DOI: [10.1109/TNS.2006.869826](https://doi.org/10.1109/TNS.2006.869826).
- [3] J. Allison et al. “Recent developments in Geant4”. In: *Nuclear Instruments and Methods in Physics Research A* 835 (2016), pp. 186–225. DOI: [10.1016/j.nima.2016.06.125](https://doi.org/10.1016/j.nima.2016.06.125).
- [4] Geant4 Collaboration. *Physics Reference Manual: Hadronic Physics*. https://geant4.web.cern.ch/documentation/pipelines/master/prm_html/PhysicsReferenceManual/hadronic/index.html. Accessed 2026-07-23. 2026.
- [5] ASIoP ZDC FastMC Project. *Independent Audit of the Legacy Recurrent HurdleGraph Model*. Internal evidence package: analysis.md. 2026.
- [6] ASIoP ZDC FastMC Project. *Independent Audit of HGF-ZDC*. Internal evidence package: analysis(1).md. 2026.
- [7] ASIoP ZDC FastMC Project. *Proposal: Single-Stage Graph Flow Baseline, Trained on the Same Data*. Internal evidence package: PROPOSAL_single_stage_flow_baseline.md. 2026.
- [8] Y. A. Kulchitsky and V. B. Vinogradov. “On the parameterization of the longitudinal hadronic shower profiles in combined calorimetry”. In: (2000). arXiv: [hep-ex/0001027](https://arxiv.org/abs/hep-ex/0001027).
- [9] G. Grindhammer and S. Peters. “The parameterized simulation of electromagnetic showers in homogeneous and sampling calorimeters”. In: *Nuclear Instruments and Methods in Physics Research A* 290 (1990), pp. 469–488. arXiv: [hep-ex/0001020](https://arxiv.org/abs/hep-ex/0001020).
- [10] M. Gallio, W. Klempt, L. Leistam, J. De Groot, and J. Schukraft. *ALICE Zero-Degree Calorimeter (ZDC): Technical Design Report*. Tech. rep. CERN-LHCC-99-005; ALICE-TDR-3. CERN, 1999. URL: <https://cds.cern.ch/record/381433>.

- [11] U. Dmitrieva and I. Pshenichnov. “On the performance of Zero Degree Calorimeters in detecting multinucleon events”. In: *Nuclear Instruments and Methods in Physics Research A* (2018). arXiv: [1805.01792](#).
- [12] M. Wojnar et al. “Machine Learning Methods for Simulating Particle Detector Responses: The ALICE Zero Degree Calorimeter Case”. In: (2023). arXiv: [2306.13606](#).
- [13] M. Kita, J. Dubinski, P. Rokita, and K. Deja. “Generative Diffusion Models for Fast Simulations of Particle Collisions at CERN”. In: (2024). arXiv: [2406.03233](#).
- [14] M. Wojnar. “Even Faster Simulations with Flow Matching: A Study of Zero Degree Calorimeter Responses”. In: (2025). arXiv: [2507.18811](#). URL: https://github.com/m-wojnar/faster_zdc.
- [15] P. Bedkowski et al. “ExpertSim: Fast Particle Detector Simulation Using Mixture-of-Generative-Experts”. In: (2025). arXiv: [2508.20991](#). URL: <https://github.com/patrick-bedkowski/expertsim-mix-of-generative-experts>.
- [16] E. Majerz, W. Dzwiniel, and J. Kitowski. “Inverse Autoregressive Flows for Zero Degree Calorimeter Fast Simulation”. In: (2025). arXiv: [2512.20346](#).
- [17] C. Krause and D. Shih. “CaloFlow: Fast and Accurate Generation of Calorimeter Showers with Normalizing Flows”. In: *Physical Review D* 107 (2023), p. 113003. arXiv: [2106.05285](#).
- [18] E. Buhmann et al. “iCaloFlow: Fast and Inductive Simulation of Calorimeter Showers with Normalizing Flows”. In: (2023). arXiv: [2305.11934](#).
- [19] E. Buhmann, S. Diefenbacher, E. Eren, G. Kasieczka, A. Korol, S. Pitz, and D. Shih. “L2LFlows: Generating High-Fidelity 3D Calorimeter Images”. In: (2023). arXiv: [2302.11594](#).
- [20] T. Buss et al. “Convolutional L2LFlows: Generating Accurate Showers in Highly Granular Calorimeters”. In: (2024). arXiv: [2405.20407](#).
- [21] L. Favaro, A. Ore, S. P. Schweitzer, and T. Plehn. “CaloDREAM—Detector Response Emulation via Attentive Flow Matching”. In: *SciPost Physics* 18 (2025), p. 088. arXiv: [2405.09629](#).
- [22] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. “Flow Matching for Generative Modeling”. In: *International Conference on Learning Representations*. 2023. arXiv: [2210.02747](#).
- [23] A. Tong, K. Fatras, N. Malkin, G. Huguet, Y. Zhang, J. Rector-Brooks, G. Wolf, and Y. Bengio. “Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport”. In: (2023). arXiv: [2302.00482](#).
- [24] S. Schnake et al. “CaloGraph: Graph-based Calorimeter Simulation”. In: (2024). arXiv: [2402.11575](#).
- [25] O. Amram and K. Pedro. “Denoising Diffusion Models with Geometry Adaptation for High Fidelity Calorimeter Simulation”. In: (2023). arXiv: [2308.03876](#).
- [26] M. Wojnar. *faster_zdc: Flow-Matching Models for ALICE ZDC Responses*. https://github.com/m-wojnar/faster_zdc. 2025.
- [27] Y. Lu, J. Collado, D. Whiteson, and P. Baldi. “SARM: Sparse Autoregressive Model for Scalable Generation of Sparse Images in Particle Physics”. In: (2020). arXiv: [2009.14017](#).
- [28] C. Krause et al. “CaloPointFlow II: Generating Calorimeter Showers as Point Clouds”. In: (2024). arXiv: [2403.15782](#).
- [29] E. Buhmann et al. “CaloClouds: Fast Geometry-independent Highly-granular Calorimeter Simulation”. In: *Journal of Instrumentation* 18 (2023), P11025. arXiv: [2305.04847](#).

- [30] W. Kool, H. van Hoof, and M. Welling. “Stochastic Beams and Where to Find Them: The Gumbel-Top-k Trick for Sampling Sequences Without Replacement”. In: *International Conference on Machine Learning*. 2019. arXiv: [1903.06059](#).
- [31] S. M. Xie and S. Ermon. “Reparameterizable Subset Sampling via Continuous Relaxations”. In: (2019). arXiv: [1901.10517](#).
- [32] A. F. T. Martins and R. F. Astudillo. “From Softmax to Sparsemax: A Sparse Model of Attention and Multi-Label Classification”. In: *International Conference on Machine Learning*. 2016. arXiv: [1602.02068](#).
- [33] B. Peters, V. Niculae, and A. F. T. Martins. “Sparse Sequence-to-Sequence Models”. In: *Annual Meeting of the Association for Computational Linguistics*. 2019. arXiv: [1905.05702](#).
- [34] C. Louizos, M. Welling, and D. P. Kingma. “Learning Sparse Neural Networks through L_0 Regularization”. In: *International Conference on Learning Representations*. 2018. arXiv: [1712.01312](#).
- [35] M. Wojnar. *zdc: Deep Generative Models for ALICE ZDC Responses*. <https://github.com/m-wojnar/zdc>. 2024.
- [36] E. Buhmann et al. “CaloClouds II: Ultra-Fast Geometry-independent Highly-granular Calorimeter Simulation”. In: (2023). arXiv: [2309.05704](#).
- [37] ATLAS Collaboration. “AtlFast3: The Next Generation of Fast Simulation in ATLAS”. In: *Computing and Software for Big Science* 6 (2022), p. 7. arXiv: [2109.02551](#).
- [38] R. Kansal et al. “Evaluating Generative Models in High Energy Physics”. In: (2022). arXiv: [2211.10295](#).
- [39] M. S. M. Sajjadi, O. Bachem, M. Lucic, O. Bousquet, and S. Gelly. “Assessing Generative Models via Precision and Recall”. In: *Advances in Neural Information Processing Systems*. 2018. arXiv: [1806.00035](#).
- [40] T. Kynkaanniemi, T. Karras, S. Laine, J. Lehtinen, and T. Aila. “Improved Precision and Recall Metric for Assessing Generative Models”. In: *Advances in Neural Information Processing Systems*. 2019. arXiv: [1904.06991](#).
- [41] T. Buss et al. “The First Full Physics Benchmark for Highly Granular Calorimeter Surrogates”. In: (2025). arXiv: [2511.17293](#).
- [42] A. Butter et al. “Generative Networks for Precision Enthusiasts”. In: (2022). Often discussed under the term calomplification. arXiv: [2202.07352](#).
- [43] C. Krause et al. “CaloChallenge 2022: A Community Challenge for Fast Calorimeter Simulation”. In: (2025). arXiv: [2410.21611](#). URL: <https://arxiv.org/abs/2410.21611>.
- [44] C. Krause and D. Shih. “CaloFlow II: Even Faster and Still Accurate Generation of Calorimeter Showers with Normalizing Flows”. In: *Physical Review D* 107 (2023), p. 113004. arXiv: [2110.11377](#).
- [45] V. Mikuni and B. Nachman. “CaloScore v2: Single-shot Calorimeter Shower Simulation with Diffusion Models”. In: *Journal of Instrumentation* 19 (2024), P02001. arXiv: [2308.03847](#).